Local Retrieval Scores Do Not Identify Faithfulness in Retrieval-Augmented Generation

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Abstract

RAG papers often assume that improving local query-passage relevance is evidence that generated answers will be more faithful. We test when such faithfulness claims are identifiable once retrieval similarity, context-set structure, evaluator choice, transfer, and cost are separated. ControlledRAG uses controlled similarity matching, a scaled SQuAD/Mistral rerun, multi-metric scoring of identical generations, cross-dataset threshold transfer, and cost-aware baselines. The results are mostly negative. At fixed mean query-passage similarity, HIGH-CCS contexts do not improve faithfulness over LOW-CCS contexts (0.636 vs. 0.639; Wilcoxon $p = 0.628$; 95% bootstrap CI $[-0.022, 0.017]$), while hallucinating more often (16.5% vs. 9.0%). Scaling the original refinement cell to $n = 500$ across five seeds reduces the DeBERTa-measured effect to 1.1 points, with only 1/5 seeds significant. The same 7,500 generations imply different effect sizes under DeBERTa, a second NLI model, and a local RAGAS-style judge (0.011, 0.032, and 0.140), with DeBERTa–RAGAS correlation $r = 0.182$. Threshold transfer is uneven, and no tested baseline is uniformly best once latency and indexing cost are included. CCS remains useful as a diagnostic: coherent uninformative noise hurts less than random noise. ControlledRAG packages these controls as a reusable protocol for auditing RAG faithfulness claims.

1 Introduction

Retrieval-augmented generation (RAG) papers often report an improvement in a retrieval score and then infer an improvement in answer faithfulness. The reasoning is understandable. If the retriever returns passages that better match the query, then the generator should have better evidence. Dense retrieval, reranking, compression, and post-retrieval refinement all inherit some version of this local premise.

The premise is incomplete. A generator receives a context set, not independent passage scores. Individually relevant passages can be redundant, mutually inconsistent, or missing the bridge needed to support an answer. Conversely, a lower-scoring passage may contain the one fact that makes the answer grounded. Faithfulness also depends on the evaluator: an NLI scorer, another NLI scorer, and an LLM judge can assign materially different support values to the same answer. A deployment claim then adds another layer: a method that improves one dataset row may be too slow, too threshold-sensitive, or worse on another task.

The central question is therefore:

> *Which RAG faithfulness claims remain identifiable once local retrieval similarity, context-set structure, evaluator choice, transfer, and cost are separated?*

We answer with a controlled falsification suite built around a tempting but ultimately rejected mechanism. The starting variable is the Context Coherence Score (CCS), a retrieval-time statistic that measures mean pairwise similarity among retrieved passages. CCS is a plausible set-level complement to query-passage relevance: it asks whether the retrieved passages look mutually compatible, rather than whether each passage looks relevant in isolation. A natural hypothesis is that this coherence variable explains why some retrieval refinements appear to reduce faithfulness.

The controlled test rejects the strong version of that story. We construct HIGH- and LOW-CCS context sets that are matched on mean query-passage similarity. The manipulation creates a large coherence gap; the faithfulness effect is null and slightly negative. This null is not a cosmetic limitation. It is the central methodological result: an intuitive RAG mechanism can look plausible under aggregate retrieval and faithfulness tables, then disappear when the relevant controls are imposed.

The rest of the paper shows that this failure is part of a wider identifiability problem. Local retrieval similarity does not identify answer support. CCS alone does not identify answer support. A single automatic metric does not identify intervention effect size. A threshold tuned on one dataset does not cleanly identify a transferable policy. A one-row baseline comparison does not identify deployment superiority once latency and indexing cost are included.

Contributions.

- **A controlled falsification of causal coherence.** We isolate context coherence from mean query similarity and show that higher CCS does not improve faithfulness in a preregistered SQuAD/Mistral intervention.
- **A scale correction of the refinement paradox.** The original pilot-sized DeBERTa effect shrinks to 1.1 points at $n = 500$ across five seeds, with only one seed reaching significance for the baseline–HCPC-v1 contrast.
- **A multi-metric non-identifiability result.** The same generations yield materially different conclusions under DeBERTa, a second NLI model, and a local RAGAS-style judge.
- **A threshold-transfer and baseline/cost audit.** We show that threshold recovery is dataset-dependent and that HCPC-v2, CRAG, RAPTOR-2L, and Self-RAG do not form a single deployment ranking.
- **A released controlled protocol and artifact.** We package matched controls, per-query generations, metric scores, threshold-transfer matrices, noise controls, and Pareto accounting as ControlledRAG.

The paper's strongest claim is deliberately narrow: standard RAG reporting is insufficient without controls that separate retrieval scores, context-set structure, evaluator choice, transfer, and cost.

2 Problem: Identifying Faithfulness

Local relevance. Let q be a query and $P = \{p_1, \ldots, p_k\}$ a retrieved context set. A standard local query-passage score measures how much each passage resembles the query, often through embedding cosine similarity. We write the mean local similarity as

$$\mathrm{Sim}_q(P) = \frac{1}{k} \sum_i \cos(e(q), e(p_i)).$$

This score is useful for retrieval, but it treats the passages as separable items.

Context coherence. The Context Coherence Score is a set-level statistic:

$$\mathrm{CCS}(P) = \frac{2}{k(k-1)} \sum_{i<j} \cos(e(p_i), e(p_j)).$$

It measures whether retrieved passages are mutually similar. CCS can expose structure that Sim_q discards, but it is not assumed to be a sufficient condition for answer support.

Faithfulness. A generator G produces $a = G(q, P)$. We use faithfulness to mean answer support by the retrieved context: the answer should be entailed by, or at least directly grounded in, P. This is a property of the set and the answer together. It is not implied by high local similarity, because a context set can look query-relevant while omitting the decisive evidence, duplicating the same partial fact, or combining passages that invite an unsupported synthesis. High CCS has the same problem in another form: topically coherent passages may all be answer-absent.

Identifiability. We use *identifiability* in the empirical sense. A reported statistic identifies a faithfulness claim only if the claimed direction would remain distinguishable after the relevant alternatives are separated. If a method has higher $\mathrm{Sim}_q(P)$ but lower support, local similarity did not identify faithfulness. If a coherence effect disappears after matching $\mathrm{Sim}_q(P)$, coherence alone did not identify the effect. If different automatic judges give different intervention magnitudes on identical generations, a single metric did not identify the effect size.

Research hypotheses. The controlled suite tests five hypotheses that are often implicit in RAG papers:

H1: Coherence causality. At fixed $\mathrm{Sim}_q(P)$, higher CCS increases answer faithfulness.

H2: Scale robustness. The refinement paradox remains large and statistically stable at larger n and multiple seeds.

H3: Metric invariance. Faithfulness conclusions are stable across reasonable automatic evaluators.

H4: Threshold transfer. A refinement policy tuned on one dataset transfers to others without large recovery loss.

H5: Method dominance. A coherence-aware policy dominates alternative RAG interventions after latency and indexing cost are included.

The results reject H1, weaken H2, reject H3, qualify H4 as dataset-dependent, and reject H5. The remaining positive claim is narrower: CCS is a diagnostic variable that can reveal structure missed by local similarity, especially in the coherent-noise control, but it is not a validated mechanism.

3 Controlled Experimental Design

The experiments are a controlled falsification suite. Each test isolates a claim that can look convincing in a conventional aggregate RAG table, then asks whether that claim survives when the relevant confounder is separated.

Pre-registration. Sample sizes, seeds, primary statistics, and decision rules were frozen before execution. Paired continuous contrasts use Wilcoxon signed-rank tests and Cohen's d_z. Means use 10,000-resample bootstrap confidence intervals. Hallucination rates use Wilson intervals. HCPC thresholds remain frozen except in the threshold-transfer experiment, where tuning and evaluation datasets are explicitly separated. No paid API calls are used in the revision results.

Test 1: separating coherence from local relevance. For each SQuAD query, we retrieve the top 20 passages and search over 3-passage subsets. We choose one HIGH-CCS and one LOW-CCS set with matched mean query similarity ($|\Delta\mathrm{Sim}_q| \leq 0.02$). This directly tests H1: whether coherence improves faithfulness after local relevance is held fixed.

Test 2: scaling the refinement effect. We rerun the SQuAD/Mistral headline cell at $n = 500$ over five seeds and three conditions: dense baseline, HCPC-v1 refinement, and HCPC-v2 gated refinement. This tests whether the original pilot-sized refinement paradox remains large and stable at scale.

Test 3: changing the faithfulness observer. The same 7,500 generations are scored with DeBERTa-v3 NLI, a second NLI model, and a local RAGAS-style judge. This keeps the model outputs fixed and varies only the observer, isolating metric dependence.

Test 4: transferring policy thresholds. We tune the HCPC threshold τ on each of five datasets and evaluate it on all five. A robust policy should not require large in-domain tuning advantage.

Test 5: separating coherence from generic noise. We compare random off-topic noise with same-topic but answer-uninformative noise. If coherence is only another name for retrieval quality, the two noise types should have similar faithfulness slopes. If coherence carries diagnostic signal beyond generic degradation, same-topic uninformative noise should be less harmful.

Test 6: comparing deployment policies. We evaluate HCPC-v2, CRAG, RAPTOR-2L, and Self-RAG on SQuAD and HotpotQA at $n = 200$ per dataset/condition. The verified full Self-RAG artifact contains 1,600 per-query evaluations, 8 summary rows, and 0 error rows. We report faithfulness, hallucination rate, mean latency, p99 latency, and indexing cost. A separate RAPTOR audit reports p50/p99 latency, dense-index cost, tree-build cost, and index size across SQuAD, PubMedQA, and HotpotQA.

3.1 ControlledRAG as a Reusable Protocol

ControlledRAG is intended as an audit protocol for RAG claims, not as a one-off set of experiments for this paper. The unit of analysis is a claim made by a RAG system paper: that a retrieval change improves support, that a threshold generalizes, that a metric identifies an effect size, or that a method is deployment-preferable. Each protocol step asks what else would need to be true for that claim to be identified.

Table 1: ControlledRAG audit controls. Each control can be reused with other datasets, retrievers, generators, and faithfulness metrics.

Control	Claim it tests	Failure mode exposed	Primary artifact/table
Similarity-matched intervention	Set coherence improves faithfulness after local relevance is fixed	A plausible set statistic can lose its apparent effect once local query–passage similarity is controlled	Table 2; `paired_wilcoxon.csv`
Multi-seed scale correction	A pilot effect remains large under repeated sampling	Small effects can be seed-sensitive even when the direction looks stable in a small headline cell	Table 3; `headline_table.csv`
Fixed-generation multi-metric scoring	A single automatic judge identifies the intervention effect size	The same generations can imply different effect magnitudes under different faithfulness observers	Table 4; `table1` multimetric CSV
Threshold-transfer matrix	One tuned threshold generalizes across datasets	In-domain tuning can hide poor off-diagonal transfer	Table 6; `tau` transfer matrix CSV
Coherent-vs-random noise separation	Coherence is just generic retrieval quality under another name	Same-topic uninformative context and random off-topic context can have different faithfulness slopes	Table 7; `slope_response.csv`
Cost-aware Pareto audit	One method is deployment-preferable	A method can win one dataset or metric while losing on p99 latency, hallucination, or offline indexing cost	Figure 2; Table 8

The protocol is deliberately modular. A future RAG paper can replace the datasets, generators, retrievers, or judges while keeping the same audit questions: match the confounder, scale the headline cell, rescore fixed generations, transfer thresholds, separate noise types, and report the cost surface.

4 Information-Theoretic Consistency Check

This section is not a proof of the empirical mechanism. Its role is narrower: it checks that the observed non-identifiability is compatible with a simple information view of retrieval.

Let Y denote the answer-supporting information needed for query q, and let each passage p_i contain two kinds of signal: local query evidence measured by a score $s(q, p_i)$, and answer-supporting information $I(Y; p_i \mid q)$ that may not be captured by the local score. A refinement rule that keeps passages or sub-passages only through $s(q, p_i)$ is using an incomplete statistic unless s is sufficient for answer support.

Proposition 1 (Incomplete local scores)**.** *If two passages can have the same local score but different $I(Y; p_i \mid q)$, then a policy that ranks only by local score can replace a more answer-informative context with a less answer-informative one while leaving the reported retrieval score unchanged or improved.*

Implication. The proposition is structural. It says that local scoring leaves room for discarding answer-relevant information; it does not say that this failure is common, large, or caused by CCS. Those are empirical questions, and the matched coherence experiment rejects the strong causal version for the primary SQuAD/Mistral setting.

Theorem 1 (Consistency of the controlled tests)**.** *Suppose a RAG claim is based on a statistic $z(q, P)$ that is not sufficient for answer support. Then there can exist two context-selection policies with the same or better value of z but different answer faithfulness. Identifying the faithfulness effect therefore requires additional controls, such as matching local similarity, varying the evaluator, checking threshold transfer, or reporting deployment cost.*

Use in this paper. The theorem motivates the experimental design, not the conclusion. The results come from the preregistered controls in Section 3. The information-theoretic view only explains why a local retrieval score can be an insufficient basis for faithfulness claims.

5 Results

5.1 H1 rejected: coherence does not causally improve faithfulness at fixed relevance

The matched construction succeeds: 200 SQuAD query pairs are built with mean absolute similarity gap 0.006 and mean CCS gap 0.533. This is a strong manipulation of set coherence while local relevance is nearly fixed. The faithfulness effect is null (Table 2).

Table 2: Matched-similarity intervention. Higher CCS does not improve faithfulness at fixed mean query similarity.

Setting	n	HIGH faith	LOW faith	HIGH–LOW	p	95% CI
SQuAD / Mistral / DeBERTa	200	0.636	0.639	-0.002	0.628	[-0.022, 0.017]

The binary hallucination result is even less compatible with a simple causal story: HIGH-CCS sets hallucinate more often than LOW-CCS sets (16.5% vs. 9.0%). We therefore reject H1. The result turns the old coherence story into evidence for the paper's current thesis: without a matched control, a plausible retrieval-set statistic can be mistaken for a faithfulness mechanism.

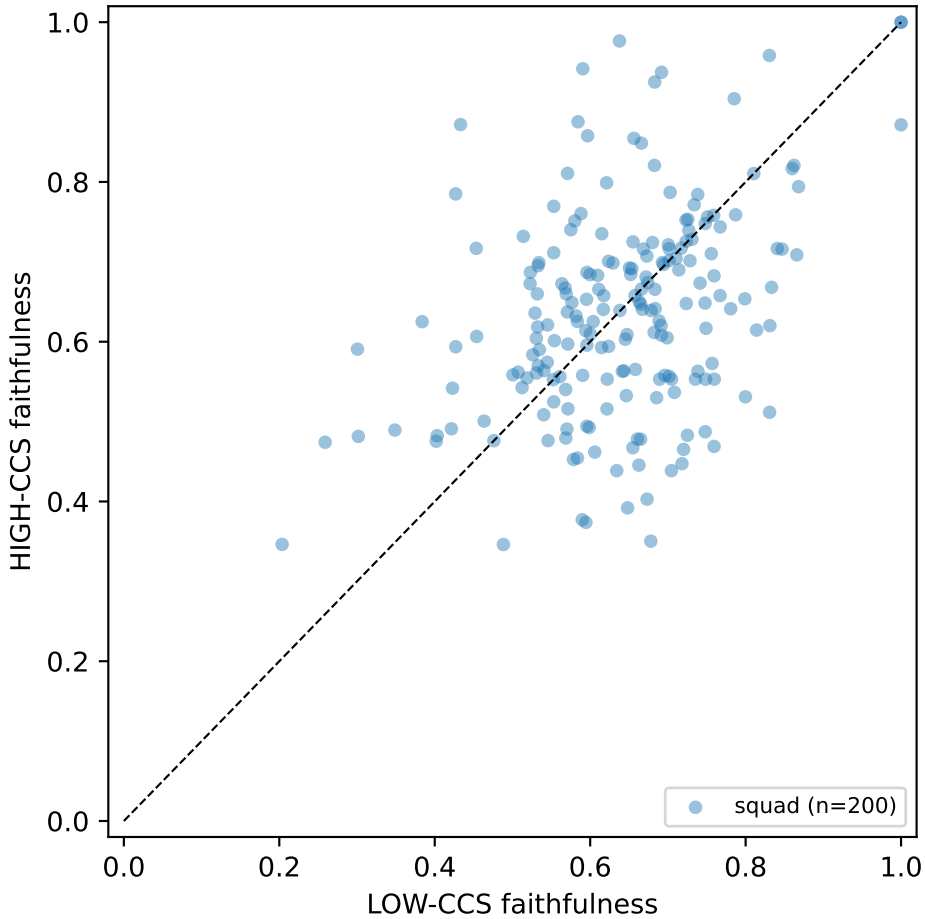

Figure 1: Paired HIGH–LOW CCS faithfulness differences under the matched-similarity intervention. The distribution is centered near zero, and the Wilcoxon test does not reject equality ($p = 0.628$), so the controlled test rejects the causal coherence hypothesis rather than supporting it.

5.2 H2 weakened: the scaled DeBERTa refinement effect is small

Scaling the SQuAD/Mistral cell to $n = 500$ over five seeds reduces the DeBERTa-measured baseline-minus-refinement effect to 1.1 faithfulness points (Table 3). Only one of five seeds is significant for the baseline-minus-HCPC-v1 contrast. HCPC-v2 restores the HCPC-v1 drop on average, but the effect is not the large, stable phenomenon suggested by the pilot.

Table 3: Scaled SQuAD/Mistral headline cell.

Condition	n	Faithfulness	Hallucination	Retrieval similarity
Baseline	2500	0.661	0.146	0.532
HCPC-v1	2500	0.650	0.147	0.569
HCPC-v2	2500	0.661	0.143	0.532

This is still an interesting non-monotonicity: HCPC-v1 raises retrieval similarity while lowering DeBERTa faithfulness. But it is a small effect under this metric, not a headline-scale failure. The original $n = 30$ result should therefore be treated as pilot evidence.

5.3 H3 rejected: the same generations imply different conclusions

Metric choice changes the magnitude of the core result (Table 4). This is the most important positive result in the revision: evaluator choice is not a reporting detail, but part of the identified estimand. The same 7,500 generations yield a 1.1 point baseline-minus-HCPC-v1 effect under DeBERTa, 3.2 points under a second NLI model, and 14.0 points under a local RAGAS-style judge. DeBERTa and RAGAS correlate only $r = 0.182$; the second NLI model and RAGAS correlate $r = 0.674$.

Table 4: Central metric-dependence result: identical generations produce different intervention magnitudes under different faithfulness judges.

Condition	n	DeBERTa	Second NLI	RAGAS-style judge
Baseline	2500	0.661	0.350	0.730
HCPC-v1	2500	0.650	0.318	0.590
HCPC-v2	2500	0.661	0.351	0.728
Baseline–v1	–	0.011	0.032	0.140

Table 5: Metric agreement on the same 7,500 generations.

Metric pair	Pearson r	Spearman ρ
DeBERTa / second NLI	0.259	0.265
DeBERTa / RAGAS-style judge	0.182	0.212
Second NLI / RAGAS-style judge	0.674	0.651

A single faithfulness metric does not identify the effect size of a RAG intervention. The result is not merely that metrics have different scales; the weak DeBERTa–RAGAS correlation means that the evaluator changes the empirical conclusion.

5.4 H4 dataset-dependent: threshold transfer is uneven

For evaluation dataset d and threshold τ, the recovery score is

$$\text{Recovery}(d, \tau) = \frac{\text{Faith}_{\text{gate}}(d, \tau) - \text{Faith}_{\text{HCPC-v1}}(d)}{\text{Faith}_{\text{baseline}}(d) - \text{Faith}_{\text{HCPC-v1}}(d)}.$$

The diagonal recovery is the recovery obtained when τ is tuned and evaluated on the same dataset. The off-diagonal mean averages recovery when that tuned τ is transferred to the other four datasets. The gap is their difference. Scores can exceed 1 when the gated policy exceeds the baseline relative to HCPC-v1, and they can be negative when the transferred threshold is worse than HCPC-v1 under this normalization.

The 5×5 threshold-transfer matrix shows large in-domain advantages for SQuAD, PubMedQA, and NaturalQuestions (Table 6). HotpotQA and TriviaQA do not show the same pattern. Thus, a single threshold tuned on one dataset cannot be treated as generally validated.

Table 6: Threshold-transfer recovery score: in-domain versus cross-dataset threshold effect.

Tune dataset	Diagonal recovery	Off-diagonal mean	Gap
PubMedQA	1.452	0.451	1.002
NaturalQuestions	1.401	0.211	1.191
SQuAD	0.823	-0.141	0.963
TriviaQA	0.464	0.698	-0.234
HotpotQA	-0.112	0.093	-0.205

5.5 Coherence remains diagnostically useful

The noise experiment prevents an overcorrection. Although CCS is not causal in Test 1, coherence is not equivalent to generic retrieval degradation. Random off-topic noise has a steeper faithfulness slope than same-topic answer-uninformative noise: -0.069 vs. -0.043 per unit noise rate (Table 7). Same-topic noise still hurts, but it hurts less.

Table 7: Noise slope response.

Condition	Faith slope	Similarity slope	Drop at full noise
Random noise	-0.069	-0.481	0.052
Coherent uninformative noise	-0.043	-0.113	0.042
HCPC-v1 refinement	–	–	0.001

This supports a narrower research claim: set-level context structure is a useful diagnostic variable, but it is neither sufficient for support nor validated as an empirical mechanism.

5.6 H5 rejected: no policy is uniformly best after cost is included

Figure 2 and Table 8 report the full SQuAD/HotpotQA baseline comparison using the verified Self-RAG artifact. HCPC-v2 is competitive on SQuAD, but RAPTOR-2L is slightly higher in faithfulness and lower in p99 latency. On HotpotQA, CRAG wins on faithfulness, hallucination rate, and latency. Self-RAG performs poorly in this matched short-answer harness.

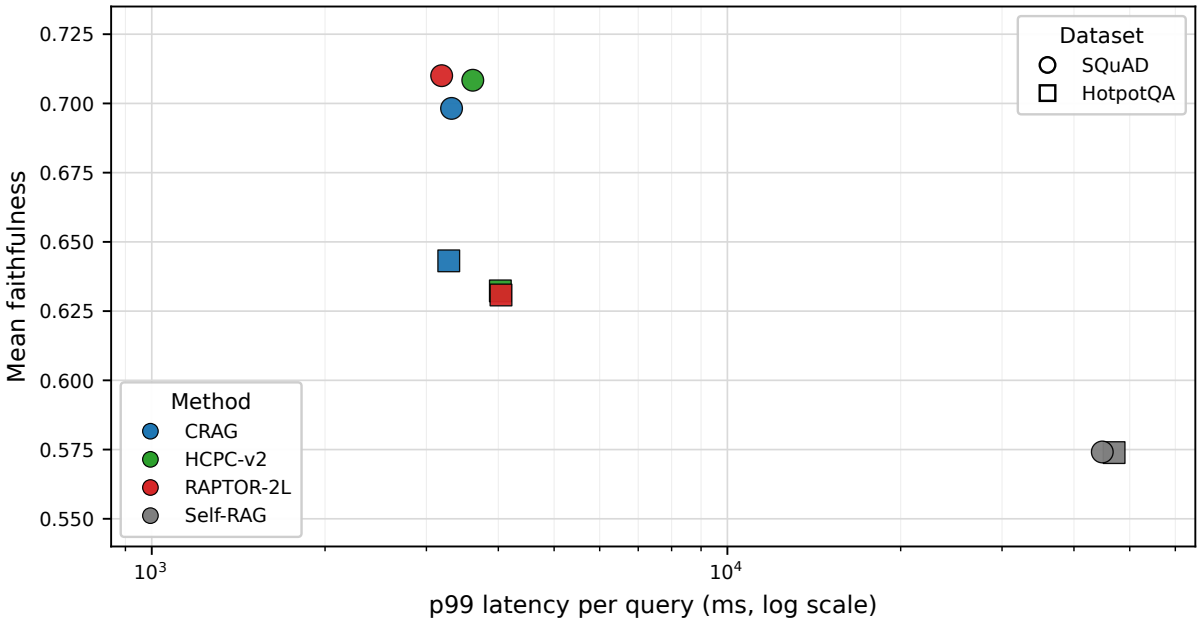

Figure 2: Cost-aware Pareto view of the full baseline audit. The x-axis uses p99 latency on a log scale so the Self-RAG tail remains visible without compressing the other methods. No method is uniformly best across datasets once faithfulness, tail latency, hallucination behavior, and offline indexing costs are considered.

Table 8: Head-to-head baseline comparison, $n = 200$ per dataset/condition. Latency is milliseconds.

Dataset	Method	Faith	Halluc.	Mean lat.	p99 lat.
SQuAD	CRAG	0.698	0.120	1021	3318
SQuAD	HCPC-v2	0.708	0.125	1349	3612
SQuAD	RAPTOR-2L	0.710	0.125	1296	3189
SQuAD	Self-RAG	0.574	0.390	7542	44800
HotpotQA	CRAG	0.643	0.105	1473	3281
HotpotQA	HCPC-v2	0.632	0.130	1841	4032
HotpotQA	RAPTOR-2L	0.631	0.130	1906	4043
HotpotQA	Self-RAG	0.574	0.430	6158	46965

RAPTOR's offline tree-building cost is also nontrivial: 99.7–161.2 seconds across datasets, compared with 1.2–8.3 seconds for dense indexing (Table 9). Method claims that omit indexing and tail latency miss part of the deployment result.

Table 9: RAPTOR cost audit across three datasets, $n = 100$ each.

Dataset	Faith	Halluc.	p50 lat.	p99 lat.	Dense idx.	RAPTOR idx.
SQuAD	0.789	0.050	1190	4943	8.333 s	161.203 s
PubMedQA	0.560	0.290	3901	6812	1.177 s	108.264 s
HotpotQA	0.617	0.210	1973	4274	7.140 s	99.740 s

6 Discussion

Standard RAG reporting is insufficient. The central result is not that one retrieval policy should replace another. It is that a common evidential pattern in RAG papers is too weak: local retrieval score improves, one faithfulness metric changes, and a deployment claim follows. Our controls show that each step can fail. Local relevance does not identify answer support; CCS alone does not identify answer support; a single automatic metric does not identify effect size; and one baseline row does not identify a deployment winner.

Why the null result matters. The matched-coherence null is not a failed paper; it is the paper's strongest control. Without it, one might conclude from correlations and pilot tables that coherence is the mechanism behind refinement failures. With it, the evidence supports a more careful interpretation: coherence is one retrieval-set property that can diagnose fragility, but support also depends on answer coverage, redundancy, contradiction, generator behavior, and evaluator bias. The collapse of the intuitive story is the point: it shows why identifiability controls are needed before RAG papers turn retrieval statistics into faithfulness claims.

What CCS is after the controls. CCS is a diagnostic variable, not a mechanism. It is useful because it measures a set-level property that local relevance misses and because the noise experiment shows that coherence-preserving degradation behaves differently from random degradation. It should not be used by itself to certify answer support.

Metric disagreement is central. The multi-metric result is not an appendix detail. A 1.1-point DeBERTa effect, a 3.2-point second-NLI effect, and a 14.0-point local RAGAS-style effect on the same generations lead to different scientific narratives. Until RAG faithfulness metrics are better calibrated against human judgments, papers should report metric disagreement rather than hide it behind a preferred scalar. In this setting, metric disagreement is part of the estimand, not a reporting nuisance.

Deployment claims need cost surfaces. The baseline audit rejects a single method ranking. RAPTOR-2L narrowly leads SQuAD faithfulness in the full head-to-head table, CRAG wins HotpotQA on faithfulness, hallucination rate, and latency, and Self-RAG underperforms in this matched short-answer harness. Offline indexing cost and p99 latency are part of the result, not engineering afterthoughts. Deployment claims require dataset-specific behavior, mean latency, p99 latency, and offline indexing cost alongside faithfulness.

Implications for RAG evaluation. The common RAG reporting pattern, "retrieval metric up, faithfulness metric up, therefore method works," is insufficient. We recommend that research papers report at least:

1. query-passage similarity and retrieval-set coherence;
2. paired query-level contrasts with bootstrap intervals;
3. multiple faithfulness metrics;
4. threshold transfer across datasets;
5. hallucination rates with Wilson intervals;
6. mean latency, p99 latency, and offline indexing cost.

7 Related Work

Retrieval-augmented generation. Retrieval-augmented generation connects parametric language models to external evidence at inference time [Lewis et al., 2020, Karpukhin et al., 2020]. The standard pipeline retrieves passages, optionally reranks or compresses them, and then conditions a generator on the selected context [Nogueira and Cho, 2019, Glass et al., 2022, Xu et al., 2024, Izacard et al., 2023]. Recent surveys document many variants of this pattern, including adaptive retrieval, query rewriting, reranking, and retrieval-time filtering [Gao et al., 2024, Chen et al., 2024]. Much of this work reports retrieval quality and answer quality side by side. That convention is useful, but it leaves a gap: a high-scoring passage set can still omit, fragment, or contradict the evidence needed to support the generated answer.

Retrieval evaluation versus answer support. Dense retrievers and cross-encoders are normally optimized for local query–passage relevance [Karpukhin et al., 2020, Reimers and Gurevych, 2019, He et al., 2021]. Benchmarks such as BEIR made cross-domain retrieval transfer a first-class evaluation problem [Thakur et al., 2021]. RAG faithfulness is a different estimand. The generator observes a set of passages, not an isolated passage, and the answer may depend on coverage, redundancy, ordering, contradiction, and whether the needed span is present at all. Work on distracting context and long-context position effects shows that extra retrieved text can change model behavior even when some relevant evidence is present [Shi et al., 2023, Liu et al., 2024]. ControlledRAG brings that distinction into the evaluation protocol by separating local similarity from set-level structure and answer support.

Faithfulness, factuality, and hallucination metrics. Automatic faithfulness evaluation has a long record of useful but imperfect proxies. NLI-style factual consistency metrics, summarization consistency benchmarks, RAGAS-style RAG metrics, and atomic factuality metrics all capture different slices of the problem [Maynez et al., 2020, Honovich et al., 2022, Laban et al., 2022, Es et al., 2024, Min et al., 2023]. Surveys of hallucination emphasize the same point: unsupported generation is not a single observable variable [Ji et al., 2023]. Our contribution is not another faithfulness metric. We hold the generations fixed and show that three plausible faithfulness observers yield materially different intervention magnitudes.

LLM-as-judge and evaluator disagreement. LLM-based judges have become common because they are flexible and inexpensive relative to full human annotation [Liu et al., 2023, Zheng et al., 2023]. They also introduce evaluator choice as a modeling decision. Prior work studies human agreement, preference benchmarks, and judge bias in open-ended generation. In RAG, the same issue appears as an identification problem: if a retrieval intervention changes DeBERTa faithfulness by 1.1 points but a local RAGAS-style judge by 14.0 points on the same generations, the intervention effect is not identified by a single metric. The disagreement is part of the claim being estimated.

Corrective and self-reflective RAG. Corrective RAG methods try to detect retrieval failure and repair the evidence before generation, while Self-RAG trains a model to retrieve, generate, and critique using reflection tokens [Yan et al., 2024, Asai et al., 2024]. These methods are natural baselines for our audit because they explicitly respond to retrieval failure rather than only changing the retriever. Our results should

not be read as a global ranking of these systems. They show that in a matched short-answer harness, policy comparisons change across SQuAD and HotpotQA and depend on hallucination rate, mean latency, p99 latency, and indexing cost.

Hierarchical retrieval and cost accounting. RAPTOR builds tree-structured summaries so retrieval can operate over both raw chunks and higher-level summaries [Sarthi et al., 2024]. That design can help when answer support is distributed across passages, but it also introduces an offline tree-building cost. Many RAG papers report online answer quality while leaving indexing cost and tail latency outside the main result table. ControlledRAG treats those quantities as part of the deployment claim. A method that improves faithfulness on one dataset but has larger p99 latency or higher offline cost has not established deployment superiority by faithfulness alone.

Benchmarks and cross-dataset transfer. Cross-dataset evaluation is standard in retrieval, but less consistently used for RAG thresholds and faithfulness interventions. BEIR-style transfer asks whether a retriever trained or selected in one domain remains effective in another [Thakur et al., 2021]. We ask an analogous question for RAG policies: does a threshold tuned to recover faithfulness on one dataset transfer to the others? The answer is uneven in our audit, with SQuAD, PubMedQA, and NaturalQuestions showing large in-domain versus off-diagonal gaps.

What prior work leaves open. Prior work evaluates RAG systems, retrieval models, and faithfulness metrics. This paper instead tests whether common RAG faithfulness claims are identifiable once local similarity, context-set coherence, evaluator choice, threshold transfer, and cost are separated. That shift is the main difference: we are not proposing a new dominant RAG policy, but a controlled protocol for deciding which claims the evidence supports.

8 Limitations and Ethics

Scope. The benchmark supports claims about short-answer extractive QA. It should not be used to claim that the same paradox, or the same policy ranking, holds for long-form synthesis, agentic workflows, or domain-specific retrieval without additional controls.

Faithfulness metrics. The multi-metric result is a strength and a limitation. It shows that evaluator choice matters, but it does not establish which evaluator is correct. No completed two-rater human calibration is reported; a 99-item template is staged for future calibration.

Thresholds and CCS. Thresholds do not transfer cleanly across datasets: SQuAD, PubMedQA, and NaturalQuestions show large in-domain versus off-diagonal recovery gaps. CCS is also not causal in the matched-similarity test and has regimes where it is uninformative about support, including coherent context sets that omit the answer. It should be treated as one diagnostic among several.

Self-RAG harness. Self-RAG underperforms in our short-answer harness. This should be reported carefully: the result is evidence about this matched inference setup, not a global claim that Self-RAG is weak. Different prompting, decoding, or task formats could change the outcome.

Compute and reproducibility. The revision was run under a zero-dollar constraint using local models and free GPU sessions. That improves accessibility but leaves one planned independent 70B reproduction blocked. The earlier 70B observation is therefore not treated as an independently rerun revision result. The artifact should disclose this plainly rather than replace it with paid or unverifiable runs.

Ethical considerations. The benchmark is designed to reduce hallucination overclaims in RAG systems, which is especially relevant for domains where unsupported answers can cause harm. The main ethical risk is misuse: a single metric or a single dataset row could be cherry-picked to market a RAG system as faithful. The benchmark therefore emphasizes per-query logs, multiple metrics, and cost-aware reporting.

9 Conclusion

This paper began from a simple suspicion: RAG systems can improve local retrieval scores while making answer support worse. The controlled evidence shows a sharper result. Faithfulness is not identifiable from local retrieval scores, not identifiable from CCS alone, not stable under a single automatic judge, and not summarized by a one-row baseline comparison.

The contribution is therefore not a new dominant RAG policy. It is a research finding about the structure of RAG evaluation: faithfulness claims require controls that current reporting often omits. Context coherence remains useful, but as a diagnostic variable inside a broader controlled evaluation protocol.

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